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CONTRIBUTION OF GROUP MEMBERS

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CHAPTER I

1.1 Overview of the Organization

Qulapong, originally known as “Q Aiskrim Gula Apong,” is a specialty ice cream brand founded by Qashidee Ajmal in 2015. The brand name is derived from the initial “Q,” representing the founder’s name and his siblings, with a secondary aspiration for customers to form queues (“queue”) for the product. The brand was officially simplified to “Qulapong” three years ago to strengthen its market identity.

The venture was born out of necessity and entrepreneurial spirit in 2015. While Qashidee is a qualified pilot (CPL) and was awaiting a career opportunity in aviation, he was encouraged by his wife to start a business. Inspired by the long queues at a popular ice cream spot in Kuching, Sarawak, he decided to enter the ice cream industry. He conducted extensive research and development (R&D) to create his own unique recipe after initial attempts were hindered by equipment issues.

The core business philosophy is built on a passion for creating genuinely addictive products. Qashidee’s approach to product development is hands-on and quality obsessed. He personally refines recipes until he himself is “addicted” to the product, considering this the ultimate test for market readiness. A blind taste test method is employed for new products, where staff sample competitors’ products (e.g., Starbucks) against his own; only his product is launched if it is consistently preferred. This demonstrates a commitment to superior taste and quality over marketing gimmicks.

The core business philosophy is built on a passion for creating genuinely addictive products. Qashidee’s approach to product development is hands-on and quality obsessed. He personally refines recipes until he himself is “addicted” to the product, considering this the ultimate test for market readiness. A blind taste test method is employed for new products, where staff sample competitors’ products (e.g., Starbucks) against his own; only his product is launched if it is consistently preferred. This demonstrates a commitment to superior taste and quality over marketing gimmicks.

The business launched successfully in Bintulu, Sarawak, in June 2015. After resolving initial technical challenges by September, the outlet saw rapid success, with daily sales growing to approximately RM 1,500 within days due to high demand from locals and visitors from nearby cities like Miri. This success led to expansion into Peninsular Malaysia with an outlet in Nilai in 2018. The brand further expanded with a "Queue Station" cafe concept, although it faced temporary closures due to the COVID-19 pandemic. The brand's resilience and unique product have been central to its sustained market presence.

1.2 Description of the Data Management System

Data collection (Qulapong's owner does all the thing alone by himself, he collects the data manually from POS and Bayarlah)	
Sales Transaction Data	Qulapong collects its data mainly through its **point-of-sale (POS) system** and manual reports from the staff. To record sales transaction, item sold, total amount, and purchase time.
Payment Data	Qulapong uses Bayarlah. So, he can collect the data of time the costumer buys and the specific amount.
Data storage (Qulapong mainly relies on manual data storage, using record books to keep sales and receipt information)	
Manual Record Keeping	Qulapong stores its business and transaction data manually using a record book to record receipts and sales information

Digital Storage	Some item data is updated and stored digitally using the Store Hub system to track inventory and sales details
Backup Storage	Qulapong keeps a security backup of essential data at the owner's home for safety
Data Security (The business practices basic data security by limiting access to a few trusted staff and keeping a home backup)	
Limited Access	Only two factory staff and the owner have access to the manual record book to protect sensitive data
Home Backup	Data is backed up at home as a precaution in case the original record is lost or damage
Staff Awareness	Training staff about data protection and privacy when handling business records

Data Analysis (Qulapong's data analysis depends on manual review and the owner's personal experience. Decisions are mostly made through observation and feedback rather than digital reports.	
Manual Analysis	The owner analyses sales and business performance manually by reviewing receipts and staff feedback
Experience-Based Prediction	Business decisions and predictions are made based on the owner's experience, observation, and feelings rather than digital tools
Feedback & Improvement	The owner revises and improves business strategies and receipts by listening to others' opinions and following current market trends

1.3 Strengths and Weaknesses Identified in the Analysis

Strengths	Types of Data Management	Weakness
Flexibility and adaptability: Fast to start which means that no need systems design, integration or lengthy setup. Furthermore, easy to change process on the fly for edge cases, pilot projects or unusual data sources. The manually database also become the active involvement of humans in the process can lead to deeper understanding data.	**Data Collection Method**	Time – consuming: It may need more process which becomes slower processing. It also longer time to insight compared with automated pipelines. Besides that, delays impede time-sensitive decision-making and lack of customers' responses. Users need to spend a lot of time searching for specific information, updating records, and preparing reports. Unlike computerized databases that allow instant access and automatic processing, manual systems require physical effort such as sorting, rewriting, and filing documents. This makes the work slower and less efficient, especially when dealing with a large amount of data.
Low initial cost: They are cost-effective due to minimal hardware, maintenance and software requirements are easy to implement without complex technical knowledge.	**Data Storage Solutions**	Storage Problem: For small businesses like Qulapong, manual data storage can be a big problem because all records are kept on paper. As the business grows, more space is needed to store files, invoices, and receipts. This is difficult for them that have limited factory space. Important documents can easily be lost or damaged, and finding old information takes a lot of time. Compared to digital storage, manual systems are less efficient, take up more space, and are harder to manage.

Back-up and original records: For some historical records, a physical, manual document like books can serve as a non-corruptible original record, unlike a digital file that may become unreadable if the application or format is lost. Simplicity: Manual systems, like paper ledger are straightforward and don't rely on complex software that could have exploitable bugs or vulnerabilities.	**Data Security Measures**	Lack of Data protection: For security, Qulapong lack of security because all the data stored on paper and just back-up in the boss's home. unfortunately, the backup also in manually. Important records such as customer information or financial details can be easily accessed, stolen, or damaged by anyone. There are no passwords or encryption to keep the data safe. Besides that, paper files can be lost due to fire, water, or human error. This makes manual databases less secure and riskier compared to computerized systems that offer better data protection.
Personal control: Qulapong make sure that every day the sales have their target. The owner always checks their sales and summarizes the result without depending on the computer software. The owner also knows where the books is kept and easily make a small adjustments or correction by hand.	**Data Analysis Techniques**	Limited analysis data tools: Qulapong does not have access to advanced tools like charts, graphs, or formulas that computer software (such as Excel, SPSS, or SQL) can provide. For example, if the owner wants to find sales trends or compare customer data over time, it must calculate everything by hand which increases the chance of human error. Manual systems also make it hard to create visual reports or identify patterns quickly. Therefore, the lack of digital tools limits how deeply and efficiently the data can be analysed.

1.4 Recommendation for Improvement

Type of Data Management	Improvement
Data Collection Method:	Implement a Comprehensive Digital POS System and Standardize Data Entry: The business should transition to a modern, integrated Point-of-Sale (POS) system (beyond just StoreHub for inventory) that can capture all sales and payment data automatically in a unified digital format. This system should be able to integrate with payment platforms (QR Payment), allowing for the real-time collection of transaction details, sales, total amounts, and customer information. Staff should be trained on standardized data entry procedures to ensure accuracy and consistency across all transactions
Data Storage Solutions:	Migrate to a Centralized Cloud-Based Database: Instead of relying on manual record books and physical backups, Qulapong should migrate all its business data like sales, inventory, ingredient stock, staff, and customer details to a cloud based Relational Database Management System (RDBMS). This provides centralized storage that is accessible from multiple locations, scalable as the business grows, and includes automatic, redundant backups. This eliminates the need for physical storage space and secures historical records against fire or loss.
Data Security Measures:	Adopt Digital Security Protocols and Access Controls: Implement strong digital security measures, including user level access controls and password protection for the database, ensuring only authorized staff can view or modify specific data. We can employ encryption for sensitive data, especially payment and customer details. Establish a formal, automated disaster recovery plan which is superior to a simple home backup. Staff training should be expanded beyond basic awareness to cover digital data security

Data Analysis Techniques	Leverage Digital Business Intelligence (BI) and Reporting Tools: Utilize the automated reporting features of the POS system to replace manual analysis. Implement Business Intelligence (BI) tools like Microsoft Excel or Power BI to generate automated reports, charts, and graphs. This allows the owner to track sales trends for example, to see product popularity, peak hours, calculate important figures like Min, Max, Avg sales and make decisions supported by data rather than solely relying on observation or feeling.

1.5 Sentence from the Quran or a hadith related to the recommendation for improvement

A relevant **Quranic verse** that aligns with the idea of continuous improvement and seeking feedback for betterment:

فَاسْأَلُوا أَهْلَ الذِّكْرِ إِنْ كُنْتُمْ لَا تَعْلَمُون

And we did not send before you (o Muhammad) except men to whom we revealed the massage. Allah commands believers to seek guidance from experts

"so ask the people of knowledge if you do not know"

The verse emphasizes the important of seeking knowledge and input from others to improved and make informed decisions, which can relate to gathering feedback to enhance services.

Additionally, **a hadith** encourages consultation and collective improvement:

Religion is sincere Advice

عَن أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ أَنَّ رَسُولَ اللَّهِ ﷺ

قَال :اَلدِّينُ النَّصِيحَةُ

قِيلَ لِمَنْ؟

قَالَ: لِلَّهِ وَلِكِتَابِهِ وَلِرَسُولِهِ وَلِأَئِمَّةِ الْمُسْلِمِينَ وَعَامَّةِ أَعْيَانِهِمْ.

Abu Huraira (RA) reported: The Messenger of Allah ﷺ said:

“Religion is sincere advice.”

They asked, “To whom?” He ﷺ replied:

“To Allah, His Book, His Messenger, the leaders of the Muslims, and the general people the religion sincere advice” (sahih muslim,52)

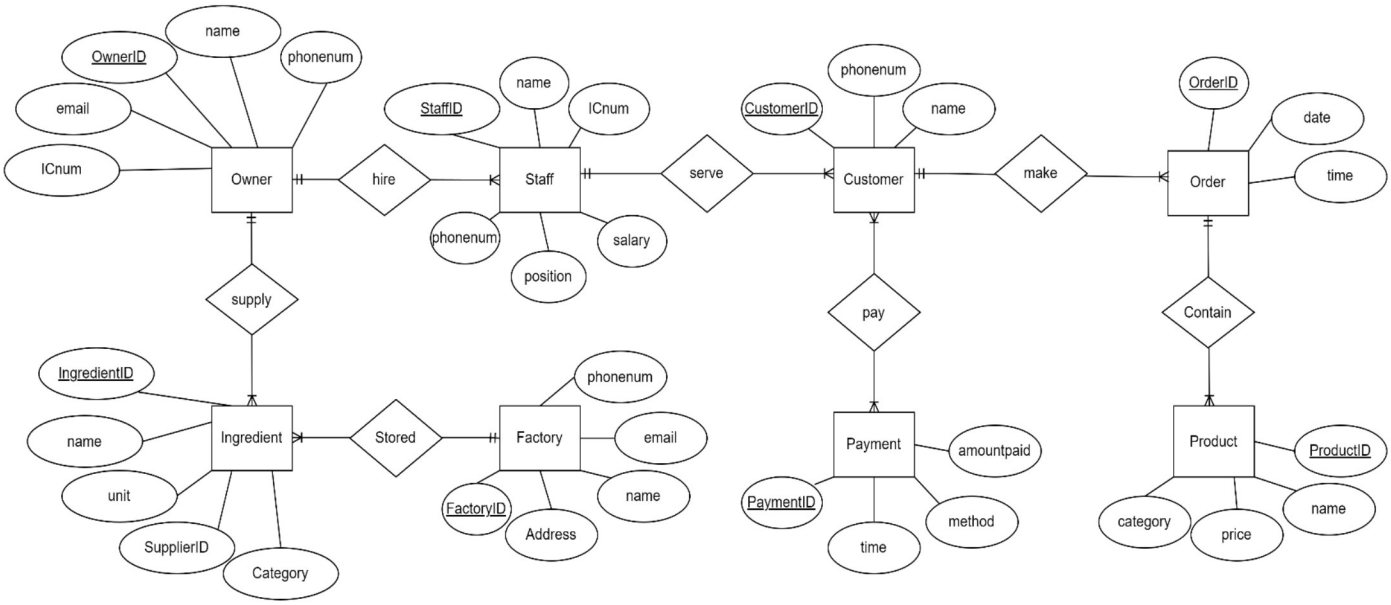
This **hadith** the value of advice and feedback as a means of growth and better service.

CHAPTER II

2.1 Business Rules

1. One or many customers pay one or many payments.
2. One owner hires one or many staff
3. One owner supply one or many ingredients.
4. One order contains one or many products.
5. One or many customers make one or many orders.
6. one or many ingredients stored one factory
7. One staff serve one or many customers

2.2 Entity Relationships Diagram



(Entity Relation Diagram)

2.3 Relational Database Schema

CUSTOMER (CustomerID,phonenum,name)

PAYMENT (PaymentID,time,method,amountpaid)

PAY (CustomerID,PaymentID)

OWNER (OwnerID,name,phonenum,email,ICnum)

STAFF (StaffID,name,ICnum,phonenum,position,salary,OwnerID)

OWNER (OwnerID,name,phonenum,email,ICnum)

INGREDIENT (IngredientID,name,unit,SupplierID,Category,OwnerID)

ORDER (OrderID,date,time)

PRODUCT (ProductID,name,price,category,OrderID)

CUSTOMER (CustomerID,phonenum,name)

ORDER (OrderID,date,time,CustomerID)

FACTORY(FactoryID,Address,name,email,phonenum)

INGREDIENT (IngredientID,name,unit,SupplierID,Category,FactoryID)

STAFF (StaffID,name,ICnum,phonenum,position,salary)

CUSTOMER (CustomerID,phonenum,name,StaffID)

CHAPTER III

3.1 Create database statement

```
mysql> CREATE DATABASE QulapongDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE QulapongDB;
Database changed
```

3.2 Create table statements

```
mysql> CREATE TABLE Owner (
    ->     OwnerID INT PRIMARY KEY,
    ->     OwnerName VARCHAR(100),
    ->     Email VARCHAR(100),
    ->     PhoneNumber VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Staff (
    ->     StaffID INT PRIMARY KEY,
    ->     StaffName VARCHAR(100),
    ->     Position VARCHAR(50),
    ->     Salary DECIMAL(10,2),
    ->     OwnerID INT,
    ->     FOREIGN KEY (OwnerID) REFERENCES Owner(OwnerID)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Customer (
    ->     CustomerID INT PRIMARY KEY,
    ->     CustomerName VARCHAR(100),
    ->     CustomerPhone VARCHAR(20),
    ->     StaffID INT,
    ->     FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Payment (
    ->     PaymentID INT PRIMARY KEY,
    ->     PaymentTime DATETIME,
    ->     PaymentMethod VARCHAR(50),
    ->     AmountPaid DECIMAL(10,2),
    ->     StaffID INT,
    ->     FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Orders (
    ->     OrderID INT PRIMARY KEY,
    ->     OrderDate DATE,
    ->     OrderTime TIME,
    ->     CustomerID INT,
    ->     PaymentID INT,
    ->     FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    ->     FOREIGN KEY (PaymentID) REFERENCES Payment(PaymentID)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Product (
    ->     ProductID INT PRIMARY KEY,
    ->     ProductName VARCHAR(100),
    ->     Price DECIMAL(10,2),
    ->     Category VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Order_Product (
    ->     OrderID INT,
    ->     ProductID INT,
    ->     PRIMARY KEY (OrderID, ProductID),
    ->     FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    ->     FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Ingredient (
    ->     IngredientID INT PRIMARY KEY,
    ->     IngredientName VARCHAR(100),
    ->     Quantity INT,
    ->     OwnerID INT,
    ->     FOREIGN KEY (OwnerID) REFERENCES Owner(OwnerID)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE Factory (
    ->     FactoryID INT PRIMARY KEY,
    ->     FactoryName VARCHAR(100),
    ->     Address VARCHAR(200)
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE Ingredient_Factory (
    ->     IngredientID INT,
    ->     FactoryID INT,
    ->     PRIMARY KEY (IngredientID, FactoryID),
    ->     FOREIGN KEY (IngredientID) REFERENCES Ingredient(IngredientID),
    ->     FOREIGN KEY (FactoryID) REFERENCES Factory(FactoryID)
    -> );
Query OK, 0 rows affected (0.01 sec)
```

Note: The OrderProduct table is created as a junction table to resolve

the many-to-many relationship between Orders and Product shown in the ERD.

3.3 Insert record statements

```
mysql> INSERT INTO Owner (OwnerID, OwnerName, Email, PhoneNumber)
    -> VALUES (1, 'Qashidee Ajmal', 'qashidee@example.com', '012-3456789');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Staff (StaffID, StaffName, Position, Salary, OwnerID)
    -> VALUES
    -> (201, 'Ali Rahman', 'Manager', 3500.00, 1),
    -> (202, 'Siti Aminah', 'Cashier', 1800.00, 1),
    -> (203, 'Ahmad Faisal', 'Chef', 2800.00, 1),
    -> (204, 'Nurul Izzah', 'Waitress', 1600.00, 1),
    -> (205, 'Hafiz Zulkifli', 'Kitchen Helper', 1500.00, 1),
    -> (206, 'Lina Abdullah', 'Supervisor', 2500.00, 1);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Customer (CustomerID, CustomerName, CustomerPhone, StaffID)
    -> VALUES
    -> (301, 'Farhan Yusof', '013-1122334', 202),
    -> (302, 'Aina Sofea', '017-5566778', 204),
    -> (303, 'Rizal Ismail', '019-8899001', 203),
    -> (304, 'Wan Amirul', '012-4455667', 206),
    -> (305, 'Diana Mariam', '018-2233445', 202);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Payment (PaymentID, PaymentTime, PaymentMethod, AmountPaid, StaffID)
    -> VALUES
    -> (801, '2025-12-10 14:30:00', 'Cash', 45.50, 202),
    -> (802, '2025-12-11 19:15:00', 'Credit Card', 78.90, 204),
    -> (803, '2025-12-12 12:45:00', 'Touch \'n Go', 32.00, 202),
    -> (804, '2025-12-13 20:10:00', 'Cash', 105.75, 206),
    -> (805, '2025-12-14 18:20:00', 'DuitNow', 63.25, 202);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Orders (OrderID, OrderDate, OrderTime, CustomerID, PaymentID)
    -> VALUES
    -> (701, '2025-12-10', '14:30:00', 301, 801),
    -> (702, '2025-12-11', '19:15:00', 302, 802),
    -> (703, '2025-12-12', '12:45:00', 303, 803),
    -> (704, '2025-12-13', '20:10:00', 304, 804),
    -> (705, '2025-12-14', '18:20:00', 305, 805);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Product (ProductID, ProductName, Price, Category)
    -> VALUES
    -> (401, 'Gula Apong Original', 6.50, 'Beverage'),
    -> (402, 'Dubai Chocolate', 12.00, 'Dessert'),
    -> (403, 'Matcha Latte', 9.00, 'Beverage'),
    -> (404, 'Espresso Original', 7.50, 'Beverage'),
    -> (405, 'Espresso Salt Caramel', 8.50, 'Beverage'),
    -> (406, 'Belgian Chocolate', 11.00, 'Dessert'),
    -> (407, 'Cold Brew', 8.00, 'Beverage'),
    -> (408, 'Chocolate Croissant', 5.50, 'Snack'),
    -> (409, 'Vanilla Cheesecake', 10.00, 'Dessert'),
    -> (410, 'Iced Matcha', 9.50, 'Beverage');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Order_Product (OrderID, ProductID)
    -> VALUES
    -> (701, 401),
    -> (701, 403),
    -> (702, 402),
    -> (702, 406),
    -> (703, 404),
    -> (703, 405),
    -> (704, 406),
    -> (704, 409),
    -> (705, 410),
    -> (705, 401);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Factory (FactoryID, FactoryName, Address)
    -> VALUES (501, 'QuLapong Factory', 'Jalan Utama, Negeri Sembilan');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Ingredient (IngredientID, IngredientName, Quantity, OwnerID)
    -> VALUES
    -> (601, 'Coffee Beans', 200, 1),
    -> (602, 'Milk', 500, 1),
    -> (603, 'Sugar', 300, 1),
    -> (604, 'Cocoa Powder', 150, 1),
    -> (605, 'Matcha Powder', 80, 1),
    -> (606, 'Caramel Syrup', 100, 1),
    -> (607, 'Salt', 250, 1),
    -> (608, 'Gula Apong Extract', 90, 1),
    -> (609, 'Ice', 1000, 1),
    -> (610, 'Whipping Cream', 120, 1);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Ingredient_Factory (IngredientID, FactoryID)
    -> VALUES
    -> (601, 501),
    -> (602, 501),
    -> (603, 501),
    -> (604, 501),
    -> (605, 501),
    -> (606, 501),
    -> (607, 501),
    -> (608, 501),
    -> (609, 501),
    -> (610, 501);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

3.4 Select statement to display records

```
mysql> SELECT * FROM Owner;
+---------+---------------+----------------------+-------------+
| OwnerID | OwnerName     | Email                | PhoneNumber |
+---------+---------------+----------------------+-------------+
|       1 | Qashidee Ajmal | qashidee@qulapong.com | 012-3456789 |
+---------+---------------+----------------------+-------------+
1 row in set (0.01 sec)
```

```
mysql> SELECT * FROM Staff;
+---------+---------------+---------------+---------+---------+
| StaffID | StaffName     | Position      | Salary  | OwnerID |
+---------+---------------+---------------+---------+---------+
|     201 | Ali Rahman    | Manager       | 3500.00 |       1 |
|     202 | Siti Aminah   | Cashier       | 1800.00 |       1 |
|     203 | Ahmad Faisal  | Chef          | 2800.00 |       1 |
|     204 | Nurul Izzah   | Waitress      | 1600.00 |       1 |
|     205 | Hafiz Zulkifli | Kitchen Helper | 1500.00 |       1 |
|     206 | Lina Abdullah | Supervisor    | 2500.00 |       1 |
+---------+---------------+---------------+---------+---------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Customer;
+------------+--------------+---------------+---------+
| CustomerID | CustomerName | CustomerPhone | StaffID |
+------------+--------------+---------------+---------+
|        301 | Farhan Yusof | 013-1122334   |     202 |
|        302 | Aina Sofea   | 017-5566778   |     204 |
|        303 | Rizal Ismail | 019-8899001   |     203 |
|        304 | Wan Amirul   | 012-4455667   |     206 |
|        305 | Diana Mariam | 018-2233445   |     202 |
+------------+--------------+---------------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Payment;
+-----------+---------------------+---------------+------------+---------+
| PaymentID | PaymentTime         | PaymentMethod | AmountPaid | StaffID |
+-----------+---------------------+---------------+------------+---------+
|       801 | 2025-12-10 14:30:00 | Cash          |      45.50 |     202 |
|       802 | 2025-12-11 19:15:00 | Credit Card   |      78.90 |     204 |
|       803 | 2025-12-12 12:45:00 | Touch 'n Go   |      32.00 |     202 |
|       804 | 2025-12-13 20:10:00 | Cash          |     105.75 |     206 |
|       805 | 2025-12-14 18:20:00 | DuitNow       |      63.25 |     202 |
+-----------+---------------------+---------------+------------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Orders;
+---------+------------+-----------+------------+-----------+
| OrderID | OrderDate  | OrderTime | CustomerID | PaymentID |
+---------+------------+-----------+------------+-----------+
|     701 | 2025-12-10 | 14:30:00  |        301 |       801 |
|     702 | 2025-12-11 | 19:15:00  |        302 |       802 |
|     703 | 2025-12-12 | 12:45:00  |        303 |       803 |
|     704 | 2025-12-13 | 20:10:00  |        304 |       804 |
|     705 | 2025-12-14 | 18:20:00  |        305 |       805 |
+---------+------------+-----------+------------+-----------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Product;
+-----------+----------------------+-------+----------+
| ProductID | ProductName          | Price | Category |
+-----------+----------------------+-------+----------+
|       401 | Gula Apong Original  |  6.50 | Beverage |
|       402 | Dubai Chocolate      | 12.00 | Dessert  |
|       403 | Matcha Latte         |  9.00 | Beverage |
|       404 | Espresso Original    |  7.50 | Beverage |
|       405 | Espresso Salt Caramel |  8.50 | Beverage |
|       406 | Belgian Chocolate    | 11.00 | Dessert  |
|       407 | Cold Brew            |  8.00 | Beverage |
|       408 | Chocolate Croissant  |  5.50 | Snack    |
|       409 | Vanilla Cheesecake   | 10.00 | Dessert  |
|       410 | Iced Matcha          |  9.50 | Beverage |
+-----------+----------------------+-------+----------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Order_Product;
+---------+-----------+
| OrderID | ProductID |
+---------+-----------+
|     701 |       401 |
|     705 |       401 |
|     702 |       402 |
|     701 |       403 |
|     703 |       404 |
|     703 |       405 |
|     702 |       406 |
|     704 |       406 |
|     704 |       409 |
|     705 |       410 |
+---------+-----------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Factory;
+-----------+-----------------+----------------------------+
| FactoryID | FactoryName     | Address                    |
+-----------+-----------------+----------------------------+
|       501 | QuLapong Factory | Jalan Utama, Negeri Sembilan |
+-----------+-----------------+----------------------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM Ingredient;
+--------------+-------------------+----------+---------+
| IngredientID | IngredientName    | Quantity | OwnerID |
+--------------+-------------------+----------+---------+
|          601 | Coffee Beans      |      200 |       1 |
|          602 | Milk              |      500 |       1 |
|          603 | Sugar             |      300 |       1 |
|          604 | Cocoa Powder      |      150 |       1 |
|          605 | Matcha Powder     |       80 |       1 |
|          606 | Caramel Syrup     |      100 |       1 |
|          607 | Salt              |      250 |       1 |
|          608 | Gula Apong Extract |       90 |       1 |
|          609 | Ice               |     1000 |       1 |
|          610 | Whipping Cream    |      120 |       1 |
+--------------+-------------------+----------+---------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Ingredient_Factory;
+--------------+-----------+
| IngredientID | FactoryID |
+--------------+-----------+
|          601 |       501 |
|          602 |       501 |
|          603 |       501 |
|          604 |       501 |
|          605 |       501 |
|          606 |       501 |
|          607 |       501 |
|          608 |       501 |
|          609 |       501 |
|          610 |       501 |
+--------------+-----------+
10 rows in set (0.01 sec)
```

3.5 Distinct code

```
mysql> SELECT DISTINCT Position
    -> FROM Staff;
+----------------+
| Position       |
+----------------+
| Manager        |
| Cashier        |
| Chef           |
| Waitress       |
| Kitchen Helper |
| Supervisor     |
+----------------+
6 rows in set (0.01 sec)

mysql> SELECT DISTINCT Salary
    -> FROM Staff;
+---------+
| Salary  |
+---------+
| 3500.00 |
| 1800.00 |
| 2800.00 |
| 1600.00 |
| 1500.00 |
| 2500.00 |
+---------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT StaffID
    -> FROM Staff;
+---------+
| StaffID |
+---------+
|     201 |
|     202 |
|     203 |
|     204 |
|     205 |
|     206 |
+---------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT Quantity
    -> FROM Ingredient;
+----------+
| Quantity |
+----------+
|      200 |
|      500 |
|      300 |
|      150 |
|       80 |
|      100 |
|      250 |
|       90 |
|     1000 |
|      120 |
+----------+
10 rows in set (0.00 sec)

mysql> SELECT DISTINCT FactoryID
    -> FROM Ingredient_Factory;
+-----------+
| FactoryID |
+-----------+
|       501 |
+-----------+
1 row in set (0.00 sec)
```

```
mysql> SELECT DISTINCT ProductID
    -> FROM Order_Product;
+-----------+
| ProductID |
+-----------+
|       401 |
|       402 |
|       403 |
|       404 |
|       405 |
|       406 |
|       409 |
|       410 |
+-----------+
8 rows in set (0.01 sec)
```

```
mysql> SELECT DISTINCT CustomerPhone
    -> FROM Customer;
+---------------+
| CustomerPhone |
+---------------+
| 013-1122334   |
| 017-5566778   |
| 019-8899001   |
| 012-4455667   |
| 018-2233445   |
+---------------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT StaffID
    -> FROM Customer;
+---------+
| StaffID |
+---------+
|     202 |
|     203 |
|     204 |
|     206 |
+---------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT PaymentTime
    -> FROM Payment;
+---------------------+
| PaymentTime         |
+---------------------+
| 2025-12-10 14:30:00 |
| 2025-12-11 19:15:00 |
| 2025-12-12 12:45:00 |
| 2025-12-13 20:10:00 |
| 2025-12-14 18:20:00 |
+---------------------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT PaymentMethod
    -> FROM Payment;
+---------------+
| PaymentMethod |
+---------------+
| Cash          |
| Credit Card   |
| Touch 'n Go   |
| DuitNow       |
+---------------+
4 rows in set (0.00 sec)

mysql> SELECT DISTINCT AmountPaid
    -> FROM Payment;
+------------+
| AmountPaid |
+------------+
|      45.50 |
|      78.90 |
|      32.00 |
|     105.75 |
|      63.25 |
+------------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT StaffID
    -> FROM Payment;
+---------+
| StaffID |
+---------+
|     202 |
|     204 |
|     206 |
+---------+
```

```
mysql> SELECT DISTINCT Price
    -> FROM Product;
+-------+
| Price |
+-------+
|  6.50 |
| 12.00 |
|  9.00 |
|  7.50 |
|  8.50 |
| 11.00 |
|  8.00 |
|  5.50 |
| 10.00 |
|  9.50 |
+-------+
10 rows in set (0.00 sec)

mysql> SELECT DISTINCT Category
    -> FROM Product;
+----------+
| Category |
+----------+
| Beverage |
| Dessert  |
| Snack    |
+----------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT DISTINCT OrderDate
    -> FROM Orders;
+------------+
| OrderDate  |
+------------+
| 2025-12-10 |
| 2025-12-11 |
| 2025-12-12 |
| 2025-12-13 |
| 2025-12-14 |
+------------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT OrderTime
    -> FROM Orders;
+-----------+
| OrderTime |
+-----------+
| 14:30:00  |
| 19:15:00  |
| 12:45:00  |
| 20:10:00  |
| 18:20:00  |
+-----------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT CustomerID
    -> FROM Orders;
+------------+
| CustomerID |
+------------+
|        301 |
|        302 |
|        303 |
|        304 |
|        305 |
+------------+
5 rows in set (0.00 sec)

mysql> SELECT DISTINCT PaymentID
    -> FROM Orders;
+-----------+
| PaymentID |
+-----------+
|       801 |
|       802 |
|       803 |
|       804 |
|       805 |
+-----------+
5 rows in set (0.00 sec)
```

3.6 Pattern Matching

```
mysql> SELECT * FROM Factory
    -> WHERE Address LIKE '%Sembilan%';
+-----------+-----------------+---------------------------------------+
| FactoryID | FactoryName     | Address                               |
+-----------+-----------------+---------------------------------------+
|       501 | QuLapong Factory | Jalan Utama, Negeri Sembilan          |
|       502 | Dessert Central  | Nilai Industrial Park, Negeri Sembilan |
|       504 | Sugar Mill       | Seremban, Negeri Sembilan             |
|       505 | Coffee Roastery  | Nilai Impian, Negeri Sembilan         |
|       507 | Packaging Dept   | Nilai, Negeri Sembilan                |
|       510 | Ice Plant        | Nilai, Negeri Sembilan                |
+-----------+-----------------+---------------------------------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Product
    -> WHERE ProductName LIKE '%Chocolate%';
+-----------+--------------------+-------+----------+
| ProductID | ProductName        | Price | Category |
+-----------+--------------------+-------+----------+
|       402 | Dubai Chocolate    | 12.00 | Dessert  |
|       406 | Belgian Chocolate  | 11.00 | Dessert  |
|       408 | Chocolate Croissant |  5.50 | Snack    |
+-----------+--------------------+-------+----------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Staff
    -> WHERE StaffName LIKE 'Ali%';
+---------+-----------+----------+---------+---------+
| StaffID | StaffName | Position | Salary  | OwnerID |
+---------+-----------+----------+---------+---------+
|     201 | Ali Rahman | Manager  | 3500.00 |       1 |
+---------+-----------+----------+---------+---------+
1 row in set (0.00 sec)
```

3.7 Min, Max & Avg

```
mysql> SELECT ROUND(MIN(Salary), 2) AS Min_Salary, ROUND(MAX(Salary), 2) AS
Max_Salary, ROUND(AVG(Salary), 2) AS Average_Salary FROM Staff;
+------------+------------+----------------+
| Min_Salary | Max_Salary | Average_Salary |
+------------+------------+----------------+
|    1500.00 |    3500.00 |        2283.33 |
+------------+------------+----------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT ROUND(MIN(Price), 2) AS Min_Price, ROUND(MAX(Price), 2) AS Max
_Price, ROUND(AVG(Price), 2) AS Average_Price FROM Product;
+-----------+-----------+---------------+
| Min_Price | Max_Price | Average_Price |
+-----------+-----------+---------------+
|      5.50 |     12.00 |          8.75 |
+-----------+-----------+---------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT MIN(Quantity) AS Min_Quantity, (Quantity) AS Max_Quantity, ROU
ND(AVG(Quantity), 0) AS Average_Quantity FROM Ingredient;
+--------------+--------------+------------------+
| Min_Quantity | Max_Quantity | Average_Quantity |
+--------------+--------------+------------------+
|           80 |          200 |              279 |
+--------------+--------------+------------------+
1 row in set (0.00 sec)
```